Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Effect on Lung Function of BDA Formulated With a Next Generation Propellant Compared With an Approved Asthma Treatment (BDA With HFA Propellant) in Participants With Asthma **Short Title/Acronym:** BDA HFO vs HFA Asthma Study **Protocol Number:** D6933C00002 **NCT Number:** NCT06502366 **Sponsor Name:** AstraZeneca **Investigational Product:** Budesonide and Albuterol (BDA) MDI HFO (Next Generation Propellant) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the pharmacodynamic (PD) equivalence of the approved asthma combination therapy, Budesonide/Albuterol (BDA), delivered using the proposed replacement propellant HFO compared with BDA delivered using the currently approved propellant HFA in participants with asthma. **Secondary Objectives:** To evaluate the overall safety, efficacy, and pharmacokinetics (AUClast and Cmax) of BDA MDI HFO compared to BDA MDI HFA and Placebo.

2.2 Background & Rationale To ensure that people with asthma can receive their combination maintenance medication effectively with a new next-generation propellant (HFO). The HFO propellant is designed to offer environmental benefits (e.g., lower global warming potential) over the currently used HFA propellant. This study is conducted to confirm that the new propellant does not compromise the clinical efficacy, lung function benefits, or safety of the BDA treatment.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active Comparator **Blinding:** Double-blind **Randomization:** Randomized (3-Way, Partial-replicate Crossover)

3.2 Planned Enrollment & Duration Target N: 422 participants **Number of Sites:** Multicenter **Estimated Study Start:** 22-Jul-2024 **Estimated Primary Completion:** 03-Mar-2026

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant must be ≥ 18 years of age at the time of signing the ICF. 2. Pre-bronchodilator FEV1 of $\ge 60\%$ and $< 90\%$ predicted normal at Visit 1 or Visit 1a. 3. Bronchodilator responsiveness defined as a $> 12\%$ and > 200 mL increase in FEV1 relative to baseline.

4.2 Exclusion Criteria 1. Confirmed or suspected diagnosis of COPD or clinically significant non-asthma airway/lung disease. 2. Systemic corticosteroid use (e.g., prednisone for 3 or more days or a single depo-injectable dose) for any respiratory, immune, or allergy-attributed disease within 6 months prior to Visit 1. 3. Hospitalization for psychiatric disorder or attempted suicide within 1 year of Visit 1.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants cross over through three regimens: BDA MDI HFO 160/180 µg, BDA MDI HFA 160/180 µg, and Placebo MDI HFA. **Route of Administration:** Oral Inhalation (Metered Dose Inhaler - MDI). **Duration of Treatment:** 3 treatment periods of 4 weeks each (Total study duration of approximately 14 to 15 weeks per participant).

5.2 Concomitant & Prohibited Therapy Permitted: Stable pre-study daily inhaled maintenance therapy with low-dose ICS (minimum 3 months) or low-dose ICS-LABA (minimum 6 months) prior to Visit 1. **Prohibited:** Systemic corticosteroids for asthma exacerbations or other immune conditions within 6 months of Visit 1 and throughout the study.

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6. Study Timeline & Visit Schedule

| Visit ID | Window | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening/Run-In** | Visits 1 & 2 (2 weeks apart) | Informed Consent, Spirometry (FEV1), Reversibility testing, Placebo run-in | | **Treatment Periods** | Visits 3, 4 & 5 (Every 4 weeks) | Crossover Randomization, Drug Administration, PK blood draws, Peak FEV1 measurements | | **End of Study** | Approx. 5 days post-final dose | Safety follow-up telephone contact, AE review, Exit assessment |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in peak Forced Expiratory Volume in 1 second (FEV1) in 0-60 minutes after dosing at Day 29. **Measurement Method:** Central/Local Spirometry.

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetics (PK): AUClast and Cmax of budesonide and albuterol. 2. Overall safety, tolerability, and clinical efficacy over the 12-week total treatment period.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical evaluations at each 4-week in-clinic visit and final safety follow-up call. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically within 24 hours of site awareness). **Data Monitoring Committee (DMC):** Managed by AstraZeneca internal safety committees to track protocol compliance and severe asthma exacerbations.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary PD equivalence outcomes; Safety Set for all adverse events. **Sample Size Justification:** Target $N=422$ powered to establish pharmacodynamic equivalence between the HFO and HFA BDA formulations in a 3-way crossover design. **Handling of Missing Data:** Mixed Model for Repeated Measures (MMRM) for continuous lung function endpoints appropriate for crossover and partial-replicate designs.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite visits and data monitoring throughout the screening and active crossover treatment periods. **Audit Trail:** eCRF locking, tracking of MDI usage, spirometry quality checks, and data clarification forms.

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11. Study Identifiers & Governance

Primary ID: D6933C00002 **Registry Number:** NCT06502366 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A Study to Investigate the Effect on Lung Function of BDA Formulated With a Next Generation Propellant Compared With an Approved Asthma Treatment (BDA With HFA Propellant) in Participants With Asthma **Official Regulatory Title:** A Randomized, Placebo-controlled, Double-blind, Multicenter, 12-Week, 3-Way, Partial-replicate Crossover Pharmacodynamic Study to Assess the Equivalence of Budesonide and Albuterol (BDA) Delivered by MDI HFO Compared With BDA Delivered by MDI HFA in Participants With Asthma

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12. Clinical Framework (Asthma Specific)

Primary Condition: Asthma **Diagnosis Threshold:** Pre-bronchodilator FEV1 of $\ge 60\%$ and $< 90\%$ predicted normal with reversibility **Medical Methodology:** Pharmacodynamic Equivalence Testing **Optimization Target:** Peak FEV1 response 0-60 minutes post-dose

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13. Study Procedures & Methodology

Management Pathway: Daily inhaled maintenance and symptom-driven therapy adjustments via crossover assignments **Education Protocol (HCP):** Site-level spirometry standardization and partial-replicate crossover compliance training **Education Protocol (Patient):** Metered Dose Inhaler (MDI) technique training and scheduled medication tracking **Quality Audit Mechanism:** Centralized spirometry reviews and strict adherence tracking to the inhaler regimens

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14. Observation & Follow-Up Schedule

Total Duration: 14 to 15 weeks **Onsite Visit Cadence:** Every 4 weeks during the active treatment crossover phases (Visits 3, 4, 5) **Remote Monitoring Frequency:** Safety follow-up telephone contact approximately 5 days after the final dose **Checklist Review Interval:** Every 2 to 4 weeks coinciding with scheduled clinic visits

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]